AFSANA KHAN

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Education

Ph.D. Nov. 2021 – Jun. 2026
***Dissertation**: "Unlocking value of Data with Vertical Federated Learning"* *Maastricht University*

M.Sc in Computer Science (Specialization: Distributed Systems) Sep. 2019 – Aug. 2021
Thesis: "Synthetic Sensor Data Generation for Authentic IoT Device Emulation" *University of Tartu*

B.Sc in Computer Science and Engineering Jan. 2014 – Jan. 2018
Thesis: "A New Algorithmic Approach to Finding Minimum Spanning Tree" *Bangladesh University of Professionals*

Experience

Researcher Nov. 2025 – Apr. 2026
Department of Advanced Computing Sciences, Maastricht University

- Worked on machine health monitoring within the DigiMach Interreg project, focusing on tool wear prediction from industrial sensor and image data.
- Developed deep learning models for high-frequency sensor streams and image data, including *CNN* architecture such as *ResNet* and investigated multimodal fusion for tool-wear prediction.
- Designed and implemented a federated learning approach to support collaborative model training across metal industry partners without sharing raw operational data.

Ph.D. Researcher Nov. 2021 – Oct. 2025
Department of Advanced Computing Sciences, Maastricht University

- Conducted a structured literature review of Vertical Federated Learning (VFL); synthesized the state of the art, mapped key challenges, and outlined future research directions.
- Designed and implemented a single-round, communication-efficient VFL method using autoencoder based compression and secure aggregation, which preserved privacy while yielding a federated model that outperformed individual local models.
- Developed a novel secure pre-training participant-selection method for VFL using secure multiparty computation (SMPC) and forward selection strategy to identify high-value parties; outperformed baseline approaches in accuracy and efficiency.
- Developed a cooperative game theory based incentive mechanism for VFL that provably ensures fair participant rewards for participating in collaboration.
- Developed a hybrid federated learning algorithm for data distributed across horizontal and vertical partitions, and applied it to suspicious financial transaction detection use case using AMLSim and financial data provided by SWIFT.
- **Provided teaching support for the Masters level courses Data Fusion and Data Mining, which included developing lab materials, delivering some lectures, and grading student assignments.**

Graduate Research Assistant Feb. 2020 – Jun. 2021
Mobile and Cloud Laboratory, Institute of Computer Science, University of Tartu

- Contributed as a Masters student to EU Horizon project *RADON* and Estonian research project C*IISIM* by engineering and deploying key technical components, including scalable cloud infrastructure on OpenStack and a proof-of-concept system for synthetic IoT data generation.

Lecturer May. 2018 – Aug. 2019
Daffodil International University

- Fulfilled a range of academic responsibilities, including delivering undergraduate lectures, supervising student projects, and organizing university workshops

Technical Skills

Programming Languages: Python (Proficient), Java (Basic)
Data Wrangling & Analytics: pandas, NumPy, Data Preprocessing, Feature Engineering, Data Visualization
Machine Learning & Modeling: Supervised Learning, Deep Learning, Multimodal Learning, Federated Learning
Libraries & Frameworks: PyTorch, scikit-learn, Flower (federated learning framework)
Tools & Infrastructure: Git, Docker, AWS (S3, Lambda), OpenStack

Publications

- **A. Khan**, M. Stallmann, M. Pietrasik, C. Kouzinopoulos, and A. Wilbik, "Federated Learning for Distributed CNC Tool Wear Prediction," *Proceedings of the 28th International Symposium on Stabilization, Safety, and Security of Distributed Systems, 2026. [Accepted]*
- **A. Khan**, M. ten Thij, G. Tang, and A. Wilbik, "HybridFL: A Federated Learning Approach for Financial Crime Detection,". *Proceedings of the 4th International Conference on Federated Learning Technologies and Applications, 2026. [Accepted]*
- **A. Khan**, M. ten Thij, G. Tang, F. Thuijsman, and A. Wilbik, "Fair Incentive Allocation in Vertical Federated Learning Using Nucleolus," in *Learning-Driven Game Theory for AI*, Elsevier, 2026.
- **A. Khan**, M. ten Thij, and A. Wilbik, "Vertical Federated Learning: A Structured Literature Review," *Knowledge and Information Systems*, Springer, Apr 2025.
- **A. Khan**, M. ten Thij, G. Tang, and A. Wilbik, "VFL-RPS: Relevant Participant Selection in Vertical Federated Learning," at International Joint Conference on Neural Networks, 2025. [*Oral Presentation*].
- **A. Khan**, M. ten Thij, F. Thuijsman, and A. Wilbik, "Using the Nucleolus for Incentive Allocation in Vertical Federated Learning," Proceedings of the 2024 2nd IEEE International Conference on Federated Learning Technologies and Applications, 2024 [*Best Paper Award*].
- **A. Khan**, M. ten Thij, and A. Wilbik, "Communication-Efficient Vertical Federated Learning," Algorithms, Volume 15, Issue 8, Article 273, 2022.
- **A. Khan**, A. Anzum, J. Sarker, S. Rahman, and J. Rahman , "An IoT-Based Intelligent Fire Evacuation System," Proceedings of the 21st International Conference of Computer and Information Technology, 2018.
- **A. Khan**, A. Anzum Aesha, and J. Sarker, "A New Algorithmic Approach to Finding Minimum Spanning Tree" Proceedings of the 1st IEEE International Conference on Electrical Engineering and ICT, 2018.

Honours and Awards

- Best Paper Award at FLTA, 2024 — 2024
- Achievement Stipend granted by University of Tartu — 2021
- European Union-funded IT Academy Specialization Stipend — 2020, 2021
- Tuition waiver scholarship granted by University of Tartu — 2019
- Bangladesh Board Scholarship for Higher Secondary School Certificate Examination — 2013

References

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Dr. Marijn ten Thij
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